8 The unit of the two electrical quantities electromotive force (e.m.f.) and potential difference (p.d.) is the volt (V).

(a) State **one** other similarity between e.m.f. and p.d.

.....

..... [1]

(b) State **one** difference between e.m.f. and p.d.

.....

..... [1]

(c) A battery consists of four cells, each of e.m.f. 1.2 V, in series.

(i) Calculate the e.m.f. of the battery.

e.m.f. = ... [1]

(ii) The battery is connected in a circuit with four 12 Ω resistors. Fig. 8.1 is the circuit diagram.

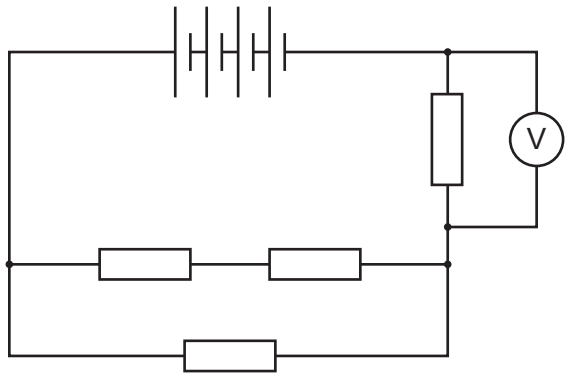


Fig. 8.1

Calculate the total resistance of this arrangement of resistors.

resistance = ... [3]

(iii) Calculate the reading on the voltmeter in Fig. 8.1.

reading = ... [2]

[Total: 8]